

BARRIERS TO EMERGENCY HEALTHCARE ACCESS IN ETHIOPIA:

UNDERSTANDING TRANSPORTATION, COMMUNICATION, AND EMERGENCY RESPONSE

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2026

ABSTRACT

Emergency healthcare access can be difficult when people cannot reach medical facilities quickly. In Ethiopia, transportation problems, road conditions, communication difficulties, waiting time, and limitations in pre-hospital emergency care can affect how patients reach hospitals. This paper reviews published research to understand some of these barriers. Studies from Addis Ababa and Jimma report important differences in ambulance use and identify factors including waiting time, distance, communication, and previous experience. Research from Addis Ababa also describes infrastructure, communication, and resource challenges in pre-hospital care. This paper considers how digital coordination might support emergency response and connects the findings to HealthReach, a student prototype developed by the author. HealthReach has not been clinically tested or validated. The purpose of this paper is to understand the problem before trying to build a solution.

Keywords: emergency healthcare, Ethiopia, ambulance, transportation, communication, pre-hospital care, digital technology

1. INTRODUCTION

The road to my church is a few kilometers long. When I was thirteen, I saw a pregnant woman being carried on a stretcher toward the nearest hospital by two men. There was no passable road for a car to reach the area. We waited to let them pass, hoping they would reach the hospital in time.

They did not.

The hospital was not far away, but reaching it was difficult. I kept thinking, “What if there was a better way?”

That experience made me interested in emergency healthcare access. However, one personal experience cannot show how common a problem is across Ethiopia. For this reason, I reviewed published research about ambulance use, transportation, waiting time, communication, and pre-hospital emergency care.

The purpose of this paper is not to claim that one technology can solve emergency healthcare problems. Instead, it examines barriers described in existing research and considers where digital coordination might be useful.

2. OBJECTIVES OF THE STUDY

The objectives of this study are:

- To identify important barriers to emergency healthcare access in Ethiopia.
 - To examine transportation and ambulance-related challenges.
 - To understand how distance, waiting time, and communication can affect emergency response.
 - To review challenges in pre-hospital emergency care.
 - To explore how digital coordination could support emergency services.
 - To connect the research findings with the HealthReach prototype.
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3. RESEARCH METHODOLOGY

This study used a literature-review method. I reviewed published research concerning emergency medical services, ambulance use, pre-hospital care, transportation, and emergency access in Ethiopia.

No original patient survey or interview was conducted for this paper, so it does not present original survey results.

The evidence comes from studies conducted in particular locations and periods. Therefore, the findings should not automatically be treated as representing every community in Ethiopia.

4. EMERGENCY MEDICAL SERVICE USE IN ADDIS ABABA

A study published in 2019 examined emergency medical service use among 429 participants who visited five selected public hospitals in Addis Ababa. The study found that 20.3% of participants arrived at the hospital by ambulance. The most common locations where emergencies occurred were homes and roadsides.

The study also examined barriers to emergency medical service use. Factors related to ambulance use included perceived waiting time, communication, and other characteristics of the emergency situation. The findings show that having an emergency medical service does not necessarily mean that every patient will use it.

This is important because emergency transportation is not only about the existence of an ambulance. People also need to know how to request help, communicate their situation, and receive a response within a reasonable amount of time.

5. AMBULANCE UTILIZATION IN JIMMA

A cross-sectional study conducted in Jimma City from June to July 2022 included 451 participants who visited emergency departments in public hospitals. The study found that ambulance services were used to bring 39.5% of the participants to hospitals.

Among the 178 patients who used an ambulance, only 35 received primary ambulance transportation directly from the emergency scene, while most arrived through inter-facility referral. The study also reported that 63 ambulance users received some form of treatment during transportation.

The researchers found that ambulance utilization was associated with several factors, including service time, referral source, mental status, first responder, distance to the hospital, and previous experience with ambulance services.

This evidence shows that ambulance access and pre-hospital treatment are connected but are not exactly the same problem. An ambulance may transport a patient, but the care available during transportation can vary.

6. CHALLENGES IN PRE-HOSPITAL EMERGENCY CARE

A qualitative study published in 2024 examined pre-hospital emergency care at the Addis Ababa Fire and Disaster Risk Management Commission. The researchers interviewed 21 experienced people working in pre-hospital emergency care.

The study identified three major areas of challenge:

1. Infrastructure
2. Communication
3. Resources

The infrastructure challenges included difficulties related to roads and other physical facilities. Communication challenges included communication gaps, problems with radio communication, and limitations involving dispatch and location information. The study also identified shortages involving professional staff, ambulances, emergency medical service centers, and other resources.

These findings are especially relevant to the question that started this project. A digital application cannot repair a road or create an ambulance. It can only support parts of the coordination process. Any technology-based solution therefore needs to be designed as support for emergency services, not as a replacement for them.

7. EMERGENCY MEDICAL SERVICES IN ETHIOPIA

A 2019 review examined literature about emergency medical services in Ethiopia from 2000 to 2018. The authors identified challenges including a shortage and unequal distribution of trained emergency medical providers, an immature emergency medical service program, limited partnerships and stakeholders, and limited motivation toward emergency medical services.

The authors concluded that long-term efforts are needed to improve emergency medical services and that a policy and analysis framework is needed to improve emergency care both inside and outside hospitals.

This broader review supports the idea that emergency healthcare is a system-level problem. Transportation is only one part of emergency response. Training, organization, communication, infrastructure, and resources also affect emergency care.

8. POTENTIAL ROLE OF DIGITAL TECHNOLOGY

Digital technology could support emergency coordination in several ways. A system could allow a person to send an emergency request, share a location, provide basic information about the situation, and notify an available ambulance driver or hospital.

Research on pre-hospital emergency care in Addis Ababa specifically identified communication problems and limitations involving geographic information systems and GPS as challenges. The researchers discussed improvements to communication systems and the use of GIS and GPS as possible areas for improving emergency response.

However, the value of a digital system would depend on the environment where it is used. It would require reliable communication networks, electricity or charged devices, accurate location information, trained users, available vehicles, and clear emergency procedures.

Technology should therefore support the existing healthcare system rather than be presented as a replacement for it.

9. HEALTHREACH PROTOTYPE

HealthReach is a student prototype I developed after thinking about the emergency healthcare problem I witnessed as a child.

The concept focuses on coordination between three main groups:

- Patients
- Ambulance drivers
- Hospitals

The goal is to explore how digital tools might make communication and emergency coordination more organized.

HealthReach is currently a prototype. It has not been clinically validated, tested in real emergencies, or evaluated by medical professionals. It should not be considered a medical device or a proven emergency service.

Further development would require professional guidance, privacy and security measures, testing, appropriate emergency protocols, and consideration of Ethiopian laws and healthcare systems.

The purpose of HealthReach at this stage is not to claim that the problem has been solved. It is to explore whether technology can help address part of the coordination problem identified through personal experience and research.

10. DISCUSSION

The evidence reviewed suggests that emergency healthcare access is an interconnected problem. Transportation, roads, distance, communication, ambulance availability, trained personnel, and pre-hospital care can all affect what happens before a patient reaches a hospital.

The studies reviewed also show that the situation can vary between locations. For example, ambulance use was 20.3% among participants in the Addis Ababa study, while the Jimma study reported ambulance use among 39.5% of its participants. These numbers come from different studies, populations, and settings, so they should not be treated as a direct comparison of the two cities.

This research changed how I think about HealthReach.

When I first thought about the problem, I mainly thought about creating a way for people to request help. After reviewing the research, I understand that a useful system would have to work within a much larger emergency-care environment.

A digital tool can help organize information and communication, but it cannot create roads, ambulances, medical staff, or hospital resources.

For me, this is an important part of learning to build technology. Before trying to solve a problem, I need to understand the problem itself.

11. LIMITATIONS

This paper is based on existing research rather than original fieldwork. The studies reviewed were conducted in particular locations and periods, so their findings cannot automatically represent all of Ethiopia.

The research also focuses mainly on selected cities and healthcare settings. Rural communities and other regions may experience different barriers that are not fully represented in the studies reviewed.

The HealthReach prototype is also conceptual and has not been tested in real emergency situations.

More research, including local field studies and professional healthcare input, would be needed before making conclusions about a digital emergency system.

12. CONCLUSION

Emergency healthcare access can be affected by many connected barriers, including transportation, distance, waiting time, communication, infrastructure, and limitations in pre-hospital emergency services.

The research reviewed in this paper shows that these challenges require more than a single solution. Emergency healthcare depends on people, vehicles, roads, communication systems, trained professionals, hospitals, and coordination between them.

This review helped me understand the problem behind the HealthReach prototype more clearly.

I started with the question, “What if there was a better way?”

Research has not given me one simple answer.

Instead, it has taught me that solving a real problem starts with understanding it.

ACKNOWLEDGEMENT

I would like to thank the researchers whose published work made this review possible. I also acknowledge the people working in emergency healthcare who deal with these challenges in real situations.

This paper was written as an independent student research project connected to my HealthReach prototype.

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